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Tesla Inc / Semiconductors | North America

Framing Tesla's Semi Manufacturing Ambitions

With AI compute emerging as a potential bottleneck, we outline what internal capacity at Tesla could look like. While any effort would be highly capital-intensive and years away from commercialization, it provides a useful framework for thinking about Tesla's longer-term AI ambitions.

Key Takeaways

- Elon Musk posted that the TSLA's Terafab project is expected to launch in 7 days. This was not previously included in its >\$20bn capex target for 2026.
- TSLA's decision to move ahead with its own chip fab is a reflection of geopolitical concerns and supply constraints to achieve LT Optimus volume targets.
- Giga Texas alone is expected to have 10m of annual humanoid production capacity, which would require 20m chips (~6x current TSLA demand for its auto business).
- If TSLA hits its long-term Optimus target (100m+ per year), it would require 200m+ chips or >50x its current demand across auto and robotaxi.

CEO Elon Musk (4Q25 Earnings Call): "So I think in order to remove the constraint, the probable constraint in three or four years, we're going to have to build a Tesla TeraFab, a very big fab that includes logic, memory and packaging domestically. And that's actually also going to be very important to ensure that we are protected against any geopolitical risks. I think people may be underweighting some of the geopolitical risks that are going to be a major factor in a few years"

Why is Tesla thinking about internal capacity? As Tesla expands its AI ambitions across FSD, robotaxi, and eventually Optimus, access to AI compute is increasingly emerging as a potential constraint. In our view, any move toward internal capacity would be aimed at supporting these in-house initiatives or related parties, rather than external entities. Based on our recent conversations with the company, its decision to move ahead with a chip fab is tied to geopolitical concerns and increasingly tied to its Optimus program, which is still several years away from scaling to meaningful volume in our view, but could potentially require substantially more chips than its auto and robotaxi business – for example, if the company hits its long-term 100m Optimus/year target, that would require 200m+ chips, which is >50x the amount of chips that it uses today in its auto/robotaxi business today.

Timing is an important consideration. Management has suggested that AI compute could become a bottleneck within the next three to four years, implying that any

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Tesla Inc (TSLA.O, TSLA US)				
Autos & Shared Mobility United States of America				
Stock Rating	Equal-weight			
Industry View	In-Line			
Price target	\$415.00			
Shr price, close (Mar 16, 2026)	\$395.56			
Mkt cap, curr (mm)	\$1,402,314			
52-Week Range	\$498.83-214.25			
Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (\$) **	1.67	1.87	2.38	3.20
Prior EPS (\$) **	-	-	-	-
ModelWare EPS (\$)	1.07	0.88	1.30	2.15
Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework				
** = Based on consensus methodology				
e = Morgan Stanley Research estimates				

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serious effort to address this constraint would need to begin well ahead of that window. As Elon Musk recently noted, removing this potential bottleneck could ultimately require building a large, integrated domestic facility encompassing logic, memory, and packaging, both to support Tesla's AI roadmap and to mitigate longer-term geopolitical risks. While Tesla's current \$20bn capex plan for 2026 does not contemplate any capex for a semi fab, **this weekend Elon posted on X that the Terafab project will launch this week, implying potential upside to capex this year.**

What would it take to get there? For context, Micron's Boise fab broke ground in September 2022 and is expected to begin shipping chips in mid-2027, underscoring the long lead times inherent in leading-edge semiconductor manufacturing.

- **WFE/Capital Outlay:** At a high level, building meaningful internal capacity would require on the order of ~100k wafers per month of leading-edge logic, implying roughly \$20-25bn of WFE. Including the cost of constructing and outfitting the facility, total capital investment could reach \$35-45bn.
- **Timing:** Tool orders - particularly from ASML - typically carry ~12-month lead times, followed by additional time for installation, qualification, and test wafer runs. Even under an aggressive scenario involving a retrofit of an existing facility, initial chip output would likely not occur until mid-2028 at the earliest. A greenfield build would extend timelines further, with meaningful output more likely 4-5 years after project initiation.

Perspectives from our semiconductor team. Even understanding Elon Musk's history of doing difficult things, this seems like a Herculean task. Advanced logic process technology nodes are built upon a foundation of prior nodes, with the steady cadence of Moore's law driven by those incremental gains. The further objective of building logic, memory, and packaging from a standing start is even more challenging - and \$20 billion plus over multiple years is probably inadequate scale to cover the development costs.

Valuation Methodology and Risks

Tesla Inc (TSLA.O)

Our PT of \$415 is comprised of 5 components: **(1)** \$45/share for core Tesla Auto business on 8.5mm units in 2040, 9.8% exit EBIT margin (ex-FSD), 10.9% WACC, and 10x 2030 exit EBITDA multiple. **(2)** Network Services at \$145/share, 80% attach rate at \$240/month ARPU by 2040. **(3)** Tesla Mobility at \$125/share on DCF with ~5mn cars at ~\$1.33/mile by 2040. **(4)** Energy at \$40/share, and **(5)** Humanoids at \$60/share (50% probability discount).

Risks to Upside

- Increased FSD attach rate / unsupervised FSD
- Cost milestones on new battery
- New model intro (Roadster, multivan)
- Robotaxi rollout
- Optimus announcement

Risks to Downside

- Competition: legacy OEMs/Chinese players/big tech in Autos and Robotics
- Execution risk across robotaxi/FSD/humanoid
- Regulatory
- China risk
- Dilution
- Valuation

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(as of February 28, 2026)

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	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1528	41%	444	49%	29%	690	42%
Equal-weight/Hold	1590	43%	374	41%	24%	725	44%
Not-Rated/Hold	4	0%	1	0%	25%	1	0%
Underweight/Sell	588	16%	87	10%	15%	220	13%
Total	3,710		906			1636	

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Stock Price, Price Target and Rating History (See Rating Definitions)

Tesla Inc (TSLA.0) - As of 03/17/26 GMT in USD
Industry : Autos & Shared Mobility



Stock Rating History: 3/1/21 : O/I; 6/22/23 : E/I; 9/10/23 : O/I; 3/13/24 : O/R; 9/25/24 : O/I; 12/7/25 : E/I

Price Target History: 2/1/21 : 293.33; 4/22/21 : 300; 10/24/21 : 400; 1/10/22 : 433.33; 6/22/22 : 400; 7/14/22 : 383.33; 8/25/22 : 383; 10/10/22 : 350; 10/24/22 : 330; 12/28/22 : 250; 1/25/23 : 220; 4/20/23 : 200; 6/22/23 : 250; 9/10/23 : 400; 10/19/23 : 380; 1/22/24 : 345; 3/5/24 : 320; 4/4/24 : 310; 12/10/24 : 400; 1/13/25 : 430; 3/20/25 : 410; 12/7/25 : 425; 1/29/26 : 415

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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INDUSTRY COVERAGE: Autos & Shared Mobility

COMPANY (TICKER)	RATING (AS OF)	PRICE* (03/16/2026)
Andrew S Percoco		
Adient PLC (ADNT.N)	E (12/07/2025)	\$19.74
Aptiv Plc (APTV.N)	E (12/07/2025)	\$71.57
Avis Budget Group Inc (CAR.O)	E (12/07/2025)	\$97.98
BorgWarner Inc. (BWA.N)	E (12/07/2025)	\$51.81
Dauch Corporation (DCH.N)	E (12/07/2025)	\$5.43
Ford Motor Company (F.N)	E (09/25/2024)	\$11.71
General Motors Company (GM.N)	O (12/07/2025)	\$72.95
Hertz Global Holdings Inc (HTZ.O)	E (02/08/2024)	\$3.97
Lear Corporation (LEA.N)	E (12/07/2025)	\$116.15
Lucid Group Inc (LCID.O)	U (12/07/2025)	\$9.89
Magna International Inc. (MGA.N)	E (09/25/2024)	\$56.06
Mobileye Global Inc (MBLY.O)	E (08/02/2024)	\$7.94
Quantumscape Corp (QS.O)	E (12/07/2025)	\$6.89
Rivian Automotive, Inc. (RIVN.O)	U (12/07/2025)	\$15.33
Tesla Inc (TSLA.O)	E (12/07/2025)	\$395.56
Visteon Corporation (VC.O)	E (06/01/2022)	\$90.06
Daniela M Haigian		
Asbury Automotive Group Inc (ABG.N)	E (09/25/2024)	\$191.00
AutoNation Inc. (AN.N)	O (09/25/2024)	\$182.78
Carmax Inc (KMX.N)	E (11/10/2025)	\$41.71
Carvana Co (CVNA.N)	O (08/07/2025)	\$305.97
Group 1 Automotive, Inc (GPI.N)	O (09/25/2024)	\$299.84
Lithia Motors Inc. (LAD.N)	E (09/25/2024)	\$252.45
Penske Automotive Group, Inc (PAG.N)	O (09/25/2024)	\$144.18
Sonic Automotive Inc (SAH.N)	E (09/25/2024)	\$60.84
Javier Martinez de Olcoz Cerdan		
Goodyear Tire & Rubber Company (GT.O)	U (09/28/2025)	\$6.74

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* Historical prices are not split adjusted.

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